Probability

Year 4 Pt 2 · **Lesson Worksheet**

Worked Examples

Probability as fraction/decimal/percentage: A bag contains 4 red, 3 blue and 5 green counters. A counter is picked at random. Write the probability it is blue as a fraction, decimal and percentage.

Answer: There are 12 counters in total, 3 of which are blue, so P(blue) = 3/12 = 1/4 = 0.25 = 25%.

Probability of NOT happening: The probability that it rains tomorrow is 0.35. What is the probability that it does NOT rain?

Answer: P(not rain) = 1 − 0.35 = 0.65.

Combined events: A fair coin is flipped twice. What is the probability of getting two heads?

Answer: List outcomes: HH, HT, TH, TT (4 equally likely outcomes). Only 1 is HH, so P(two heads) = 1/4.

Probability Basics (15 questions)

1. A fair six-sided dice is rolled once. What is the probability of rolling a 4? Give your answer as a fraction.
2. A spinner has 8 equal sections numbered 1 to 8. What is the probability of landing on an odd number?
3. Write the probability 'certain' as a number. A) 0 B) 0.25 C) 0.5 D) 0.75 E) 1
4. A bag contains 10 balls, 6 of which are yellow. What is the probability of picking a yellow ball, as a decimal?
5. Order these words from least likely to most likely: likely, impossible, evens, certain, unlikely.
6. A coin is flipped once. What is the probability of landing on tails, as a percentage?
7. Which word best describes an event with probability 0.5? A) Impossible B) Unlikely C) Evens D) Likely E) Certain
8. A bag contains only blue counters. A counter is picked at random. What is the probability it is blue?
9. List all the possible outcomes when a single coin is flipped.
10. A box contains 3 red pens and 7 blue pens. What is the probability of picking a red pen, as a fraction in its simplest form?
11. Select ALL that apply: which of these are valid probabilities for an event? A) 0 B) 1.2 C) 0.5 D) −0.3 E) 1
12. A fair six-sided dice is rolled once. What is the probability of rolling a number greater than 4? A) 1/6 B) 1/3 C) 1/2 D) 2/3 E) 5/6
13. The probability that a train is late is 3/20. Write this probability as a percentage.
14. List all the possible outcomes when a single fair six-sided dice is rolled.
15. A spinner is divided into 4 equal sections coloured red, blue, green and yellow. What is the probability of NOT landing on red?

Applying Probability (15 questions)

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1. A bag contains 5 green, 3 red and 2 yellow counters. A counter is picked at random. What is the probability it is NOT green? Give your answer as a fraction.

2. A spinner has 10 equal sections: 4 are blue, 6 are white. What is the probability of landing on white, as a decimal?

3. The probability that Amir wins a game is 0.4. What is the probability that he loses (assuming no draws)? A) 0.2 B) 0.4 C) 0.5 D) 0.6 E) 0.8

4. A fair coin is flipped twice. List all 4 possible outcomes, then state the probability of getting exactly one head and one tail.

5. A bag contains 20 sweets: 8 are orange, the rest are lemon. What is the probability of picking a lemon sweet, as a percentage?

6. Two fair coins are flipped. What is the probability of getting at least one head? A) 1/4 B) 1/2 C) 3/4 D) 1 E) 0

7. A bag has 4 red and 6 blue counters. A counter is picked, its colour noted, and it is NOT put back. If the first counter picked is red, what is the probability the second counter picked is also red?

8. Select ALL that apply: a fair six-sided dice is rolled once. Which statements are true? A) P(rolling a 7) = 0. B) P(rolling an even number) = 1/2. C) P(rolling a number less than 7) = 1. D) P(rolling a 1) = 1/6. E) P(rolling a multiple of 5) = 1/2.

9. A box of 15 chocolates has 9 milk and 6 dark chocolates. What is the probability of picking a milk chocolate, as a fraction in its simplest form?

10. A weather forecaster says there is a 70% chance of sunshine tomorrow. What is the probability, as a fraction, that it will NOT be sunny?

11. A bag contains only red and blue counters. P(red) = 0.3. If there are 40 counters in total, how many are blue?

12. A fair spinner has sections numbered 1 to 5. It is spun twice. What is the probability that it lands on 5 both times? A) 1/25 B) 1/10 C) 2/5 D) 1/5 E) 2/25

13. A drawer contains 6 black socks and 4 white socks. One sock is pulled out at random, not put back, and a second sock is pulled out. What is the probability that both socks are black?

14. In a raffle, 200 tickets are sold and Priya buys 5 of them. What is the probability that Priya's ticket is drawn as the winner, as a decimal?

15. Select ALL that apply: a bag contains 3 red, 4 blue and 3 green counters (10 total). Which statements are true? A) P(red) = 3/10. B) P(not blue) = 3/5. C) P(green) = P(blue). D) P(blue) = 0.4. E) P(red or green) = 0.6.

Probability Challenge (15 questions)

1. A bag contains red, blue and green counters only. P(red) = 1/3 and P(blue) = 1/4. If there are 24 counters in total, how many are green?

2. Two fair dice are rolled together and their scores are added. What is the probability that the total is 7? Give your answer as a fraction. A) 1/12 B) 1/9 C) 1/6 D) 1/4 E) 5/36

3. A bag has 5 red and 3 blue counters. A counter is picked at random and NOT replaced, then a second counter is picked. What is the probability that both counters are blue?

4. A spinner numbered 1–6 is spun once and a fair coin is flipped once. What is the probability of spinning an even number AND flipping heads?

5. Select ALL that apply: a box contains 12 cards numbered 1 to 12, and one card is drawn at random. Which statements are true? A) P(drawing a multiple of 3) = 1/3. B) P(drawing a square number) = 1/4. C) P(drawing a prime number) = 5/12. D) P(drawing a number greater than 10) = 1/6. E) P(drawing an even number) = 1/3.

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6. A bag contains 4 yellow and 6 purple counters. Two counters are picked one after another without replacement. What is the probability that the first is yellow and the second is purple? A) 4/15 B) 2/5 C) 8/45 D) 1/15 E) 4/9

7. The probability that it snows on a given day in January is 1/5. In a town where this probability applies independently each day, what is the probability that it snows on two consecutive days?

8. A fair six-sided dice is rolled twice. What is the probability that the second roll is a higher number than the first roll? (Hint: consider all 36 equally likely outcomes.)

9. A bag contains only red and green counters in the ratio 2:3. A counter is removed and found to be red; it is not replaced. If there were 20 counters originally, what is the probability that the next counter picked is green?

10. Select ALL that apply: two fair coins are flipped together. Which statements are true? A) There are 4 equally likely outcomes. B) P(two tails) = 1/4. C) P(at least one tail) = 3/4. D) P(exactly one head) = 1/2. E) P(no heads) = 1/2.

11. A spinner has sections 1, 2, 3, 4 and 5, each equally likely. It is spun twice and the two numbers are multiplied together. What is the probability that the product is an even number?

12. A bag contains 3 red, 2 blue and 1 green counter. A counter is picked at random and NOT replaced, then a second counter is picked. What is the probability that the two counters picked are different colours? A) 2/5 B) 3/5 C) 11/15 D) 4/5 E) 13/15

13. In a class of 30 pupils, the probability that a randomly chosen pupil has a pet is 0.6. Of those with a pet, half have a dog. What is the probability that a randomly chosen pupil has a dog?

14. A game show host has 3 identical-looking boxes; exactly one contains a prize. A contestant picks a box at random. What is the probability they do NOT pick the prize box?

15. Two spinners, each with sections 1, 2 and 3, are spun together and the scores are added. What is the probability the total is greater than 4?

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